



JOMO KENYATTA UNIVERSITY OF AGRICULTURE & TECHNOLOGY

**SCHOOL OF OPEN, DISTANCE &
eLEARNING**

**IN COLLABORATION WITH
SCHOOL OF COMPUTING AND
INFORMATION TECHNOLOGY**

**DEPARTMENT OF INFORMATION
TECHNOLOGY**

BIT 2324 GEOGRAPHICAL INFORMATION SYSTEMS

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**P.O. Box 62000, 00200
Nairobi, Kenya**

BIT 2324 GEOGRAPHICAL INFORMATION SYSTEMS

Course description

Introduction: Defining GIS and Spatial Data File Formats. The Basics of GIS Software Packages. Projections and Coordinate Systems. Coordinate Systems and Map Projections. Tabular Data Design, Functions, Pitfalls and GeoProcessing. Working with Tables, Queries, and Basic Geoprocessing Tools. Data sources and data collection – where to get data and how to create your own. Creating and editing new spatial data files in ArcMap and ArcPad. Cartographic design. Basics of Remote Sensing. Examples of GIS applications. Implementation issues and the future of GIS. Students to be given work on research projects.

Prerequisite: ICS 2206: Introduction to Database Systems

Course aims

Students will explore the theory and application of geographic information systems (GIS) by examining the general principles of GIS and practical experience in its use.

Learning outcomes

At the end of this course, you should be able to;

1. Describe the basic components of map making
2. Use GIS software and perform map projections and distortions
3. Use metadata, documentation, and instructions to solve simple technical problems, decide what kind of map they need to make, and evaluate the quality of data from various sources.
4. Evaluate other people's maps.
5. Manipulate large databases and present the material graphically in clear, readable maps.

Instruction methodology

Lectures, practical sessions in a Computer Laboratory, Assignments, Exercises and Project Work

Course Text Books

1. GIS Fundamentals: A first Text on Geographical Information Systems. Paul Bolstad. ISBN 978-0-9717647-3-6.
2. Aronoff, Stan. 1989. Geographic Information Systems: A Management Perspective, WDL Publications, Ottawa. ISBN-13: 978-0921804918
3. Harmon (2003). The Design and Implementation of Geographic Information Systems. 1st Edition. ISBN-13: 978-0471204886

Reference Textbooks

1. Demers. 2004. Fundamentals of Geographic Information Systems. ISBN-13: 978- 0471204916
2. Chang, Kang-tsung. 2010. Introduction to Geographic Information Systems, 5th ed., McGraw-Hill Higher Education, Toronto. ISBN-13: 978-0077805401
3. Campbell, Wyne. 2011. Introduction to Remote Sensing, Fifth Edition. ISBN-13: 978- 1609181765

Course Journals

1. International Journal of Geographical Information Science. ISSN 1365-8816
2. Journal of Geographical Information Systems. ISSN Online: 2151-1969
3. URISA Journal. ISSN 1045-8077 Acta Informatica ISSN 0001-5903

Reference Journals

1. Transactions in GIS. ISSN: 1361-1682
2. Journal of Geographical Systems. ISSN: 1435-5949
3. Transactions in GIS. ISSN: 1467-9671

Assessment information

The module will be assessed as follows;

- 30% Continuous Assessments (written CAT to be administered at JKUAT main campus or one of the approved centres 10%, Assignments 10%, Practical 10%)
- 70% End of Semester written Examination to be administered at JKUAT main campus or one of the approved centres

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LESSON 1

Introduction

Learning outcomes

Upon completing this lesson, you should be able to:

- *Define GIS*
- *Describe components of a GIS*
- *Distinguish different data types*

1.1. Introduction

GIS is an information system which is geographic in nature. It is a computer-based tool for mapping and analyzing geographic phenomenon that exist, and events that occur on Earth. It is a means of storing, retrieving, sorting, and comparing spatial data to support some analytic process. GIS technology integrates common database operations such as query and statistical analysis with the unique visualization and geographic analysis benefits offered by maps. It links graphical features (entities) to tabular data (attributes). These abilities distinguish GIS from other information systems and make it valuable to a wide range of public and private enterprises for explaining events, predicting outcomes, and planning strategies. GIS performs tasks faster and with more sophistication than traditional methods. GIS is a collection of computer hardware and software.

A GIS can be made up of a variety of software and hardware tools and geographic data for capturing, managing, analyzing, and displaying forms of geographically referenced information. The important factor is the level of integration of these tools to provide a smooth operating and fully functional geographic data processing environment by manipulating and managing digital spatial (geographic) and related attribute data.

In general, a GIS provides facilities for data capture, data management, data manipulation and data analysis, and the presentation of results in both graphic and report form, with a particular emphasis upon preserving and utilizing inherent characteristics of spatial data. The ability to incorporate spatial data, manage it, analyze it,

and answer spatial questions is the distinctive characteristic of geographic information systems. Geographic location is an important attribute of activities, policies, strategies, and plans. Geographic problems involve an aspect of location, either in the information used to solve them, or in the solutions themselves.

As hardware, a GIS consists of a computer, memory, storage devices, scanners, printers, global positioning system units, and other physical components. If the computer is situated on a network, the network can also be considered an integral component of the GIS because it enables sharing of data and information that the GIS uses as input and creates as outputs.

As a tool, a GIS permits maintaining, analyzing, and sharing wealth of data and information. From the relatively simple task of mapping the path of a hurricane to the more complex task of determining the most efficient garbage collection routes in a city, a GIS is used across the public and private sectors. Online and mobile mapping, navigation, and location-based services are also personalizing and democratizing GISs by bringing maps and mapping to the masses.

1.1.1. Advantages of GIS

- GIS explores both geographical and thematic components of data in a holistic way
- It stresses geographical aspects of a research question
- It allows handling and exploration of large volumes of data
- It allows integration of data from widely disparate sources
- It allows analysis of data to explicitly incorporate location
- It allows a wide variety of forms of visualization.

1.1.2. Disadvantages of GIS

- Geographical data is expensive
- The learning curve on GIS software can be long
- GIS shows spatial relationships but does not provide absolute solutions

- GIS originated from the earth sciences and computer science. Solutions may not be appropriate for humanities research

1.2. Approaches to GIS

In addition to recognizing the many definitions of a GIS, there are three general and overlapping approaches to understanding GISs

1. the application approach,
2. the developer approach,
3. the science approach.

Though most GIS users would probably identify with one approach more than another, they are not mutually exclusive. Moreover, as GISs and, more generally, information technology advance, the above approaches will be transformed and re-shaped accordingly.

The application approach to GISs considers a GIS primarily to be a tool. From this perspective, a GIS is used to answer questions, support decision making, maintain an inventory of geographic data and information, and make maps. As a tool, there are certain skills that should be acquired and required in order to use and apply a GIS properly. The application approach to a GIS is more concerned with using and applying GISs to solve problems than the GIS itself.

In light of the rapid rate of technological and GIS innovation, in conjunction with the widespread application of GISs, new questions about GIS technology and its use are continually emerging. One of the most discussed topics concerns privacy, and in particular, what is referred to as locational privacy. Questions concerning locational privacy are of relatively little concern. However, with the advent of GPS and its integration into cars and other mobile devices, questions, debates, and even lawsuits concerning locational privacy and who has the right to such information are rapidly emerging.

1.2.1. GIS Examples

- ArcView
- ArcGIS

- MapInfo

1.3. Characteristics of GIS

- GIS integrates spatial data and other kinds of information within one system: it offers a consistent framework for analyzing space
- GIS makes connections between activities based on spatial proximity
- GIS provides the mechanisms for undertaking the manipulation and display of geographic knowledge

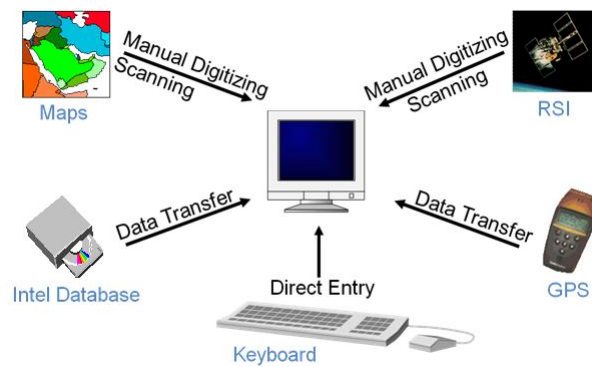
1.4. Functions/Subsystems of GIS

A GIS has four main functional subsystems. These are:

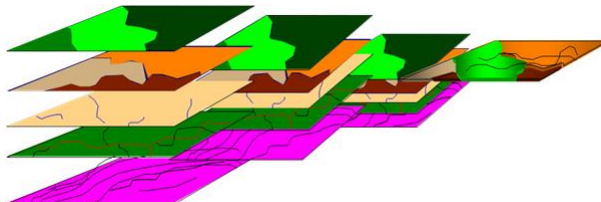
- a data input subsystem;
- a data storage and retrieval subsystem;
- a data manipulation and analysis subsystem;
- a data output and display subsystem.

1. **Data Input Assembly:** Before geographic data can be used in a GIS, the data must be converted into a suitable digital format. The process of converting data from paper maps into computer files is called digitizing. Modern GIS technology can automate this process fully for large projects using scanning technology; smaller jobs may require some manual digitizing. Today many types of geographic data already exist in GIS-compatible formats. Standard digital data exists in useable formats and can usually be loaded directly into a GIS.

A data input subsystem allows the user to capture, collect, and transform spatial and thematic data into digital form. The data inputs are usually derived from a combination of hard copy maps, aerial photographs, remotely sensed images, reports, survey documents, etc.



2. **Data Storage:** The data storage and retrieval subsystem organizes the data, spatial and attributes, in a form which permits it to be quickly retrieved by the user for analysis, and permits rapid and accurate updates to be made to the database. This component usually involves use of a database management system (DBMS) for maintaining attribute data. Spatial data is usually encoded and maintained in a proprietary file format. For small GIS projects it may be sufficient to store geographic information as simple files. There comes a point, however, when data volumes become large and the number of data users becomes more than a few, that it is best to use a database.
3. **Data Analysis and Manipulation:** This is one the most powerful functions in a GIS. It gives the user the ability to place the cartographic representation of thematic data over another. The data manipulation and analysis subsystem allows the user to define and execute spatial and attribute procedures to generate derived information. This subsystem is commonly thought of as the heart of a GIS, and usually distinguishes it from other database information systems and computer-aided drafting (CAD) systems. It is likely that data types required for a particular GIS project will need to be transformed or manipulated in some way to make them compatible with your system. GIS technology offers many tools for manipulating spatial data and for weeding out unnecessary data.



4. **Data Output:** There are many forms of output for a GIS. The first is usually a soft copy. A full functioning GIS can do this. It can plot/print given the equipment, the 2D and 3D perspective views on a 2D map. It entails Tables, Maps, Interactive Displays and 3-D Perspective View. The data output subsystem allows the user to generate graphic displays, normally maps, and tabular reports representing derived information products.

1.5. GIS Process

The GIS process involves six steps that are common to what is referred to as the end to end map-making process.

1. Define the spatial problem/question.
2. Define the GIS criteria.
3. Import or create the data sets
4. Perform the GIS analysis
5. Create the output
6. Decide whether or not the output solves or answers the spatial problem/question.
If not, then refine the problem and start the process again.



1.6. Components of a GIS

An operational GIS has a series of components that combine to make the system work. These components are critical to a successful GIS. A working GIS integrates five key components:

1. **Hardware:** Hardware is the computer system on which a GIS operates. Today, GIS software runs on a wide range of hardware types, from centralized computer servers to desktop computers used in stand-alone or networked configurations.
2. **Software:** GIS software provides the functions and tools needed to store, analyze, and display geographic information.
3. **Data:** The most important component of a GIS is the data. Geographic data and related tabular data can be collected in-house, compiled to custom specifications and requirements, or occasionally purchased from a commercial data provider. A GIS can integrate spatial data with other existing data resources, often stored in a corporate DBMS. The integration of spatial data (often proprietary to the GIS software), and tabular data stored in a DBMS is a key functionality afforded by GIS.
4. **People:** GIS technology is of limited value without the people who manage the system and develop plans for applying it to real world problems. GIS users range from technical specialists who design and maintain the system to those who use it to help them perform their everyday work. The identification of GIS specialists versus end users is often critical to the proper implementation of GIS technology.
5. **Methods:** A successful GIS operates according to a well-designed implementation plan and business rules, which are the models and operating practices unique to each organization.

As in all organizations dealing with sophisticated technology, new tools can only be used effectively if they are properly integrated into the entire business strategy and operation. To do this properly, it requires not only the necessary investments in hardware and software, but also in the retraining and/or hiring of personnel to utilize the new technology in the proper organizational context. Failure to implement your GIS without regard for a proper organizational commitment will result in an unsuccessful system.

1.7. GIS Data Types

The basic data type in a GIS reflects traditional data found on a map. Accordingly, GIS technology utilizes two basic types of data which are stored for each item in the database. These are:

1. **Spatial data:** This data describes the absolute and relative location of geographic features. It says where the feature is co-ordinate based; Vector data and discrete features which are :
 - Points
 - Lines
 - Polygons (zones or areas): – Raster data:
 - A continuous surface
2. **Attribute data:** This data describes characteristics of the spatial features. These characteristics can be quantitative and/or qualitative in nature. Attribute data is often referred to as tabular data. Eg. statistics, text, images, sound, etc.

The coordinate location of a forestry stand would be spatial data, while the characteristics of that forestry stand, e.g. cover group, dominant species, crown closure, height, etc., would be attribute data. Other data types, in particular image and multimedia data, are becoming more prevalent with changing technology. Depending on the specific content of the data, image data may be considered either spatial, e.g. photographs, animation, movies, etc., or attribute, e.g. sound, descriptions, narration's, etc.

1.8. Classification of GIS

Different classification schemes of commercial GIS are proposed in literature literature. It is convenient to classify the main GIS software packages into six groups, based on their functionality and type:

1. **Professional:** Refers to fully featured software that has the capabilities of collection and editing GIS data database administration advanced and editing GIS data , database administration , advanced geoprocessing and analy-

sis, and other specialist tools. Examples of professional GIS include ESRI ArcInfo and Smallworld GIS.

2. **Desktop:** This category is based on the desktop computer, with the focus on data use rather than data creation. It has excellent tools for making maps, reports, and charts. Most of the desktop GIS software have all adopted the Microsoft standards for interoperability and user interface. Examples include Autodesk World, ESRI ArcView, Intergraph GeoMedia, and MapInfo Professional, Quantum GIS
3. **Hand-held:** Hand-held GIS are lightweight systems designed for mobile and field use. Have capabilities similar to hand held computers, supporting many display, query, and simple analytical applications, even on displays of 320 X 240 pixels displays of 320 X 240 pixels
4. **Component:** These are basically tool kits of GIS functions (components) that a reasonably knowledgeable programmer can use to build a full GIS software system. These type of components can be used to create highly customized and optimized applications that can either be stand-alone or can be embedded within other software systems alone or can be embedded within other software systems. Examples include ESRI MapObjects, MapInfo MapX, and Blue Mable Geographics GeoObjects.
5. **Viewer:** It is a free GIS software that is able to display and query popular file formats. The GIS viewers have limited functional capabilities, restricted to display query and simple mapping display query and simple mapping. They do not support editing, sophisticated analysis, modelling or customization. Examples include ESRI's ArcReader, ArcExplorer now ArcGIS Explorer, Intergraph's GeoMedia Viewer and Mapinfo's ProViewer. The rationale of the viewers is to establish market share, and specific vendor terminology and data formats as de facto standards
6. **Internet:** These products harness the unique characteristics of the world wide by integrating GIS technology with web browsers and servers, and uses the hypertext transmission protocol (http) for communication. Examples of Internet GIS products include Autodesk MapGuide,

ESRI ArcIMS, Intergraph GeoMedia Web Map and MapInfo MapXtreme.
ArcIMS, Intergraph GeoMedia Web Map and MapInfo MapXtreme.

Three other types of GIS software systems are: CAD-based GIS, Raster-based GIS, and GIS application servers.

1.9. Spatial Data

Spatial data refers to the real-world geographic objects of interest, such as streets, buildings, lakes, and countries, and their respective locations. In addition to location, each of these objects also possesses certain traits of interest, or attributes, such as a name, number of stories, depth, or population. GIS software keeps track of both the spatial and attribute data and permits us to link the two types of data together to create information and facilitate analysis.

Spatial data is essentially data with a location. It contains information about the location and shape of, and relationship among geographic features usually stored as coordinates. Spatial data comes in two types, VECTOR and RASTER. Geographic information systems work with two fundamentally different types of geographic models—the "vector model" and the "raster model."

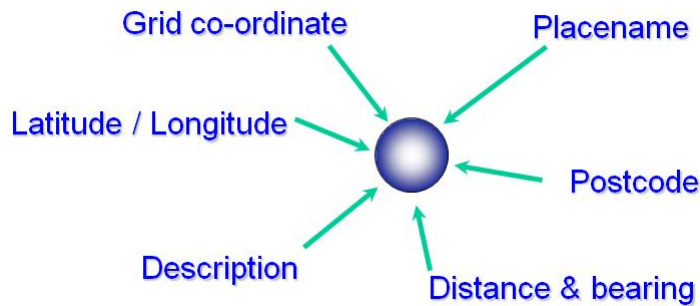
1.9.1. What makes data spatial?

Spatial data has particular characteristics. These can be described in terms of: shape, place and relationship to other spatial data (or geometry, location, and topology. It is also necessary to model real world data (such as a road or building) in terms of a geographical representation. For example, a road could be represented as a line and the building perhaps as a small box on a map. These features (line, box) are in fact models of the actual real world features. Sometimes these models are described as objects or entities too.

Another important aspect of spatial data is that it often contains attribute information. That implies that a description of the feature (the road) is held in some form. The description might be the name or the type of road (A, B,). This information might be held in a database record or simply written or depicted on a map.

Finally spatial data by its very nature implies that relationships are also recorded. When we look at map data, we automatically interpret the relative locations of the spatial data. Computers require more explicit descriptions.

Spatial data thus refers to information that is associated with a location or place. It may be recorded on a map, held as records in a database or even be represented as a photograph. Remember that Geography is, in fact, the study of spatial information and that we are surrounded by geography. You will also discover that most information is either spatial or has a spatial component.



1.10. Characteristics of spatial data

1. Location: This can be expressed in many ways. It may, however, differ in terms of:
 - Method e.g. co-ordinate, place name, Post Code
 - Precision e.g. <1.0m, City name, Post Code sector
 - Accuracy e.g. centre of building, front door position
2. Geometry: The shape of a building or county, the course of a river, the route of a road and the shape of the landscape / relief
3. Topology: This term expresses how geographical data about one location relates to data about another. These concepts often include descriptions such as:
 - Connected to
 - Within
 - Adjacent to
 - North of . . .

1.10.1. Examples of Spatial Data

- **Socio-economic data:** It is often from national and local government, and is usually the product of population surveys and censuses. This data is also used by a number of commercial vendors who combine census information with other datasets to produce neighbourhood profiles classifying particular areas for marketing purposes. This ability to recognize particular markets based on geographical datasets is known as Geodemographics and is one of the fastest growth areas within GIS.
- **Environmental Data:** The collection and analysis of information about the environment was one of the driving forces behind the development of GIS and continues to be an important application area. Environmental data sets often tend to be large and require considerable management. Data often includes boundaries between vegetation types, for example, which are fuzzy i.e. they are not defined by a simple line. Conversely, socio-economic data is usually related to administrative boundaries, which are sharp if artificial.

Revision Questions

EXERCISE 1.  Define GIS.

Example . Describe the characteristics of a GIS that distinguish it from other kinds of computerized systems.

Solution: Revise.



EXERCISE 2.  Describe your current location in terms of name of building, Post-code, Grid reference & Latitude/Longitude.

LESSON 2

Basics of GIS Software Packages

Learning outcomes

By the end of this topic you should be able to;

- *Explain GIS software*
- *Distinguish between thin and thick client servers*

2.1. Introduction

There are a number of Geographical Information Systems available. They range from high-powered analytical software to visual web applications, and each of those are used for a different purpose. Due to the vast number of GIS available, they have common features. Understanding these basic features will give you confidence with any GIS system that you use. GIS Applications are normally programs with a graphical user interface that can be manipulated using the mouse and keyboard. The application provides menus near the top of the window which, when clicked using the mouse, show a panel of actions. These actions provide a way for you to tell the GIS Application what you want to do. The software encompasses not only the GIS package, but all the software used for databases, drawings, statistics, and imaging and the functionality of the software used to manage the GIS determines the type of problems that the GIS may be used to solve. The software used must match the needs and skills of the end user. A common function of GIS Applications is to display map layers. Map layers are stored as files on a disk or as records in a database. Normally each map layer will represent something in the real world. Three groups of GIS will be covered:

- Web-based GIS: ONS and London Profiler
- Geobrowser: Google Earth
- Desktop GIS: ArcGIS

2.2. Desktop GIS

A GIS software allows you to interactively work with spatial data. A desktop GIS is a mapping software that is installed onto and runs on a personal computer. It is software for managing information of any kind according to where it is located. For example, businesses can track customer locations, optimize delivery routes, or decide where to site future businesses using GIS; scientists use GIS to manage sensitive wildlife habitats or track animal movements in an ecosystem; and health care specialists track the spread of infectious disease with GIS. GIS technology powers solutions for telecommunications, utilities, agriculture, defense, oil, health care, transportation, mining, environmental management, petroleum, water/waste-water, and many other industries as well as local, state, and federal government agencies.

Desktop GIS represents the real world on a computer similar to the way maps represent the world on paper. Both GIS and paper maps convey information about places. However, desktop GIS has power and flexibility that paper maps lack. The scale of the map influences the size of what appears on it. With GIS, however, you can store and link huge amounts of information about the objects represented on maps. These objects are called features. Each map feature has a location, a representative shape, and a symbol that represents one or more of its characteristics. Because features on maps are organized according to relative location or position, maps are good for showing the relationships among feature locations. These relationships, called spatial relationships, are important because understanding them helps us solve problems.

Maps use three basic shapes— points, lines and areas to represent real-world objects. Points represent objects that have discrete locations and are too small to be depicted as areas. Lines represent objects that have length but are too narrow to be depicted as areas. Areas represent objects too large to be depicted as points or lines. Shapes alone do not give you enough information, so maps use graphic symbols to help identify features and provide information about them. Most features can be represented as more than one shape. The scale of a map tells how the size of the map features compares with the size of the geographic objects they represent. Map scales vary from small-scale to large-scale.

With desktop GIS, you are not limited to the amount of information you can get about what you see on the map. Desktop GIS stores all the information about map features in a GIS database and links the features on the map to the information about them. This means that you can access all the information about a feature by simply clicking on it.

The information that a desktop GIS stores about map features is referred to as attribute information, or attributes. The attributes of a river, for example, might include its name, length, average depth, rate of flow, water quality, how many dams are on it, and how many bridges cross it. Desktop GIS formats attributes in rows and columns, and stores them as tables. Each column stores a different attribute and each row relates to a single feature. The link between map features and their attributes is the basic principle behind how a desktop GIS works, and is the source of its power. Once the map features and attributes are linked, you can access the attributes for any map feature or locate any feature from its attributes in a table.

Desktop GIS links sets of features and their attributes and manages them together in units called themes. A theme contains a set of related features, such as roads, streams, parcels, or wildlife habitat areas, along with the attributes for those features.

2.3. Geobrowser

A Geobrowser is better explained with reference to an internet browser, i.e. Internet Explorer. In short, a geobrowser can be understood as an Internet Explorer for geographic information. Like the internet it allows the combination of many types of geographic data from many different sources. The biggest difference between the World Wide Web and the geographic web however is that everything within the latter is spatially referenced. Google Earth is the most popular geobrowser available.

A Geobrowser is a client in a web browser or a stand-alone application that is capable of accessing georeferenced data and providing multi-dimensional visualization to end users. It aims at improving a user's experience by providing better maps, faster services and high resolution data. One of the strengths is its ability to provide content and processing power for other web applications. It has the following capabilities:

- It has a robust search capability
- It has real time monitoring and access to historical data
- It provides simple access to data collected by different scientific disciplines through the internet
- It has built-in tools for importing and exporting data

2.4. Web-based GIS

Web GIS is a mechanism to deliver maps and GIS services on the internet. Web-based GIS, or WebGIS, are online GIS applications are excellent data visualization tools. Their functionality is limited compared to software stored on the computer, but they are user-friendly and particularly useful as they not required data download. It is a viable tool for building various geospatial Web application. It uses a centralized paradigm approach. It is a cost effective, easy and efficient way to access and

disseminate the geographic information, spatial analysis tools and geospatial web services. It evolves from different web maps and can be modeled as client - server architecture or thin client architecture. Web GIS is similar to the client/server typical three-tier architecture.

2.4.1. Thin Client Server Approach

1. Advantages of thin client server approach:

- Central Control
- Easy for data version
- Generally cheaper
- Integration possibilities
- Regarding some cartographic aspects such font

2. Disadvantages of thin client server approach

- Large data volume,
- Slow response time,
- Less interactive, and
- No local accountability

2.4.2. Thick Client Server Approach

1. advantages:

- Document/graphics standards are not required
- Vector data can be used
- Image quality not restricted to GIF and JPEG
- Modern interface is possible ; it is not restricted to single – click operations .

2. Disadvantages

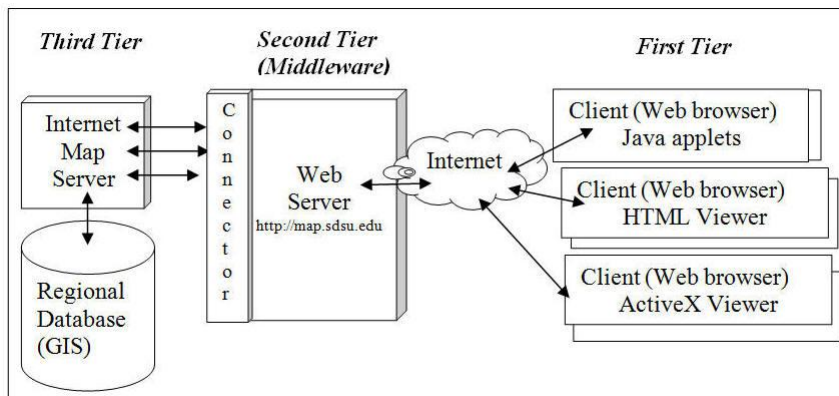
- Additional software requirement, and b) Platform/ browser incompatibility

2.4.3. Web GIS Architectures

The Web-GIS prototype implements two parts:

1. A (server-side) Web Feature Service (WFS) which interacts with clients and processes queries, retrieves and stores data to DBMS and generates WFS/OGC-conformal query responses.
2. A client, implemented as a JavaScript Class, is provided to query data, generate WFS requests and visualise geodata. The client implements WFS and WMS interfaces and can therefore be used independently from the prototype's server-side implementation.

The geo- processing breaks down into server-side and client-side tasks. A client typically is a Web browser. The server-side consists of a Web Server, Web GIS software and Database .



Revision Questions

EXERCISE 3.  Describe the different software used in GIS.

Example . Distinguish the different types of client servers.

Solution: Revise.



EXERCISE 4.  What are the characteristics of Geobrowser?

LESSON 3

Projections & Coordinate Systems

Learning outcomes

By the end of this topic you should be able to;

- *Define coordinate systems*
- *Describe different types of maps*
- *Distinguish vector, raster & image data formats*

3.1. Introduction

A map is a graphical representation of where features are, explicitly and relative to one another. It is composed of different geographic features represented as either points, lines, and/or areas. Each feature is defined both by its location in space and by its characteristics. Quite simply, a map is a model of the real world. The map legend is the key linking the attributes to the geographic features. Attributes, e.g. such as the species for a forest stand, are typically represented graphically by use of different symbology and color.

Maps and mapping are essential components of any and all geographic information systems (GISs). For instance, maps constitute both the input and output of a GIS. Hence a GIS utilizes many concepts and themes from cartography, the formal study of maps and mapping. In order to become proficient with GISs, there is need to learn more about cartography, maps, and mapping.

3.2. Map Scale, Coordinate Systems, and Map Projections

The central purpose of a map is to provide relevant and useful information to the map user. In order for a map to be of value, it must convey information effectively and efficiently. Mapping conventions facilitate the delivery of information in such a manner by recognizing and managing the expectations of map users. Mapping or cartographic conventions refer to the accepted rules, norms, and practices behind the making of maps. One of the most recognized mapping conventions is that “north is up” on most maps. Though this may not always be the case, many map users expect north to be oriented or to coincide with the top edge of a map or viewing device like a computer monitor.

All map users and map viewers have certain expectations about what is contained on a map. Such expectations are formed and learned from previous experience by working with maps. It is important to note that such expectations also change with increased exposure to maps. Understanding and meeting the expectations of map viewers is a challenging but necessary task because such expectations provide a starting point for the creation of any map.

Several other formal and informal mapping conventions and characteristics, many of which are taken for granted, can be identified. Among the most important cartographic considerations are map scale, coordinate systems, and map projections. Map scale is concerned with reducing geographical features of interest to manage-

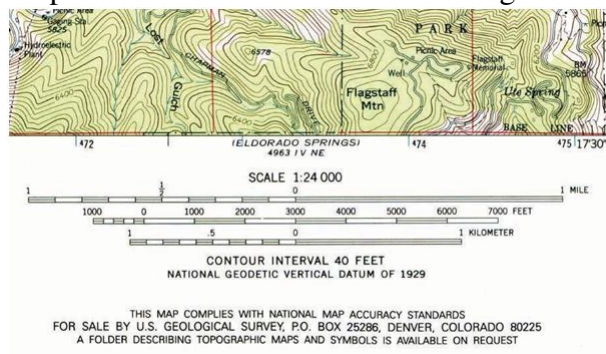
able proportions, coordinate systems help us define the positions of features on the surface of the earth, and map projections are concerned with moving from the three-dimensional world to the two dimensions of a flat map or display.

3.2.1. Map Scale

One of the challenges behind mapping the world and its resident features, patterns, and processes is reducing it to a manageable size. All maps reduce or shrink the world and its geographic features of interest by some factor. Map scale refers to the factor of reduction of the world so that it fits on a map.

Map scale can be represented by text, a graphic, or some combination of the two. For example, it is common to see “one inch represents one kilometer” or something similar written on a map to give map users an idea of the scale of the map. Map scale can also be portrayed graphically with what is called a scale bar. Scale bars are usually used on reference maps and allow map users to approximate distances between locations and features on a map, as well as to get an overall idea of the scale of the map.

Map Scale from a United States Geological Survey (USGS) Topographic Map



The representative fraction (RF) describes scale as a simple ratio. The numerator, which is always set to one (i.e., 1), denotes map distance and the denominator denotes ground or “real-world” distance. One of the benefits of using a representative fraction to describe scale is that it is unit neutral. Any unit of measure can be used to interpret the map scale. Consider a map with an RF of 1:10,000. This means that one unit on the map represents 10,000 units on the ground. Such units could be inches, centimeters, or even pencil lengths; it really does not matter.

Map scales can also be described as either “small” or “large.” Such descriptions are usually made in reference to representative fractions and the amount of detail represented on a map. For instance, a map with an RF of 1:1,000 is considered a large-scale map when compared to a map with an RF of 1:1,000,000 (i.e., $1:1,000 > 1:1,000,000$). Furthermore, while the large-scale map shows more detail and less area, the small-scale map shows more area but less detail. Clearly, determining the thresholds for small- or large-scale maps is largely a judgment call.

All maps possess a scale, whether it is formally expressed or not. Though some say that online maps and GISs are “scaleless” because we can zoom in and out at will, it is probably more accurate to say that GISs and related mapping technology are multiscalar. Understanding map scale and its overall impact on how the earth and its features are represented is a critical part of both map making and GISs.

3.2.2. Coordinate Systems

Just as all maps have a map scale, all maps have locations, too. Coordinate systems are frameworks that are used to define unique positions. For instance, in geometry we use x (horizontal) and y (vertical) coordinates to define points on a two-dimensional plane. The coordinate system that is most commonly used to define locations on the three-dimensional earth is called the geographic coordinate system (GCS), and it is based on a sphere or spheroid. A spheroid (a.k.a. ellipsoid) is simply a sphere that is slightly wider than it is tall and approximates more closely the true shape of the earth. Spheres are commonly used as models of the earth for simplicity.

The unit of measure in the GCS is degrees, and locations are defined by their respective latitude and longitude within the GCS. Latitude is measured relative to the equator at zero degrees, with maxima of either ninety degrees north at the North Pole or ninety degrees south at the South Pole. Longitude is measured relative to the prime meridian at zero degrees, with maxima of 180 degrees west or 180 degrees east.

Note that latitude and longitude can be expressed in degrees-minutes-seconds (DMS) or in decimal degrees (DD). When using decimal degrees, latitudes above the equator and longitudes east of the prime meridian are positive, and latitudes below the equator and longitudes west of the prime meridian are negative (see the following table for examples).

Converting from DMS to DD is a relatively straightforward exercise. For example, since there are sixty minutes in one degree, we can convert $118^{\circ} 15$ minutes to 118.25 ($118 + 15/60$). Note that an online search of the term “coordinate conversion” will return several coordinate conversion tools.

When we want to map things like mountains, rivers, streets, and buildings, we need to define how the lines of latitude and longitude will be oriented and positioned on the sphere. A datum serves this purpose and specifies exactly the orientation and origins of the lines of latitude and longitude relative to the center of the earth or spheroid.

Depending on the need, situation, and location, there are several datums to choose from. For instance, local datums try to match closely the spheroid to the earth’s surface in a local area and return accurate local coordinates. A common local datum used in the United States is called NAD83 (i.e., North American Datum of 1983). For locations in the United States and Canada, NAD83 returns relatively accurate positions, but positional accuracy deteriorates when outside of North America.

The global WGS84 datum (i.e., World Geodetic System of 1984) uses the center of the earth as the origin of the GCS and is used for defining locations across the globe. Because the datum uses the center of the earth as its origin, locational measurements tend to be more consistent regardless where they are obtained on the earth, though they may be less accurate than those returned by a local datum. Note that switching between datums will alter the coordinates (i.e., latitude and longitude) for all locations of interest.

For geographic data, often referred to as spatial data, features are usually referenced in a coordinate system that models a location on the earth’s surface. The coordinate system may be of a variety of types. For natural resource applications the most common are geographic coordinates such as latitude and longitude, e.g. $56^{\circ}27'40''$ and $116^{\circ}11'25''$. These are usually referred by degrees, minutes, and seconds. Geographic coordinates can also be identified as decimal degrees, e.g. 54.65° .

Geographic data is distinguished from attribute data in that it is referenced spatially by a coordinate system, e.g. it has a spatial extent. Natural resource applications commonly use a Legal Survey system.

● Entity Representations

We need symbolize spatial features in order to be able to associate attribute information. We can classify different features into different dimensions. Each classification of dimension is a conceptual classification.

- **Points** - “0” dimensionality. No length or Width.: Each point is Discrete in that it can only occupy a given point in space at any given time.
- **Lines** - “1” dimensional. Length, but No Width: Must have a beginning and an ending point.
- **Polygons** - “2” dimensional. Length and Width: By adding Width, we can describe a feature as having an area.
- **Surfaces** - “3” dimensional. Length, Width, and Height: Surfaces have infinite number of values (e.g. Elevation). We say that this type of data is Continuous.

When thinking of spatial elements, we must consider Spatial Scale. Depending on scale, we may want to represent a river as a line or a polygon.

We typically represent objects in space as three distinct spatial elements:

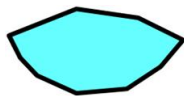
1. Points - simplest element



2. Lines (arcs) - set of connected points



3. Polygons - set of connected lines

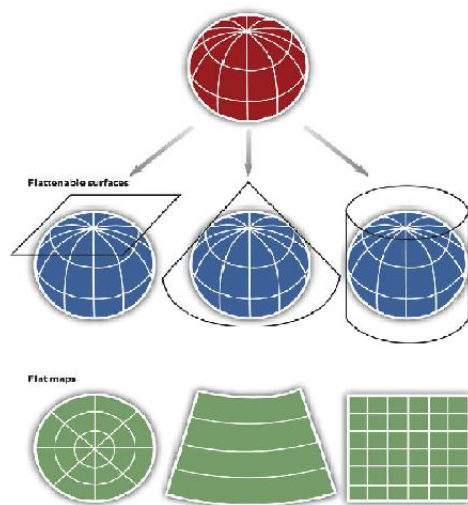


3.2.3. Map Projections

Map projections refer to the methods and procedures that are used to transform the spherical three-dimensional earth into two-dimensional planar surfaces. Specifically, map projections are mathematical formulas that are used to translate latitude

and longitude on the surface of the earth to x and y coordinates on a plane. Since there are an infinite number of ways this translation can be performed, there are an infinite number of map projections. The mathematics behind map projections are beyond the scope of this unit. To illustrate the concept of a map projection, imagine that we place a light bulb in the center of a translucent globe. On the globe are outlines of the continents and the lines of longitude and latitude called the graticule. When we turn the light bulb on, the outline of the continents and the graticule will be “projected” as shadows on the wall, ceiling, or any other nearby surface. This is what is meant by map “projection.”

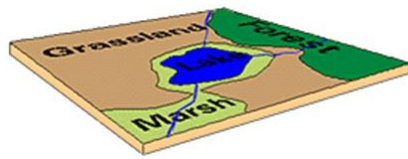
Within the realm of maps and mapping, there are three surfaces used for map projections (i.e., surfaces on which we project the shadows of the graticule). These surfaces are the plane, the cylinder, and the cone. Referring again to the previous example of a light bulb in the center of a globe, note that during the projection process, we can situate each surface in any number of ways. For example, surfaces can be tangential to the globe along the equator or poles, they can pass through or intersect the surface, and they can be oriented at any number of angles.



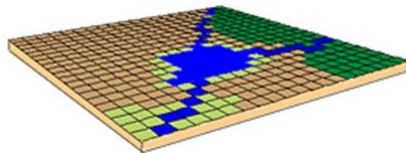
3.3. Spatial Data Models

Traditionally spatial data has been stored and presented in the form of a map. Three basic types of spatial data models have evolved for storing geographic data digitally. These are referred to as:

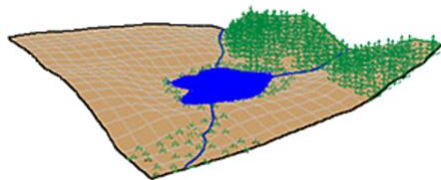
1. Vector;



2. Raster;



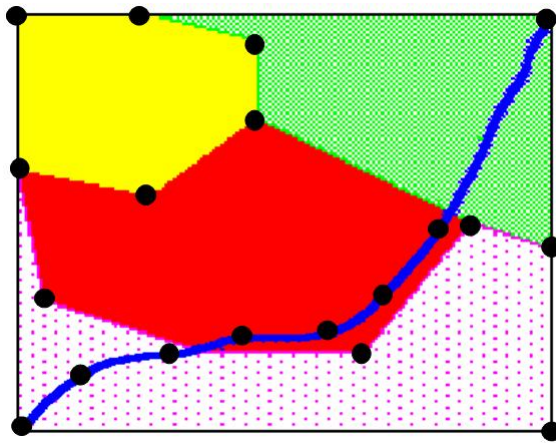
3. Image



The below diagrams reflect the two primary spatial data encoding techniques. These are vector and raster. Image data utilizes techniques very similar to raster data, however typically lacks the internal formats required for analysis and modeling of the data. Images reflect pictures or photographs of the landscape.

1. **Vector data formats**

All spatial data models are approaches for storing the spatial location of geographic features in a database. Vector storage implies the use of vectors (directional lines) to represent a geographic feature. Vector data is characterized by the use of sequential points or vertices to define a linear segment. Each vertex consists of an X coordinate and a Y coordinate.



Vector lines are often referred to as arcs and consist of a string of vertices terminated by a node. A node is defined as a vertex that starts or ends an arc segment. Point features are defined by one coordinate pair, a vertex. Polygonal features are defined by a set of closed coordinate pairs. In vector representation, the storage of the vertices for each feature is important, as well as the connectivity between features, e.g. the sharing of common vertices where features connect.

Several different vector data models exist, however only two are commonly used in GIS data storage. The topologic data structure is often referred to as an intelligent data structure because spatial relationships between geographic features are easily derived when using them. Primarily for this reason the topologic model is the dominant vector data structure currently used in GIS technology. Many of the complex data analysis functions cannot effectively be undertaken without a topologic vector data structure.

The secondary vector data structure that is common among GIS software is the computer aided drafting (CAD) data structure. This structure consists of listing elements, not features, defined by strings of vertices, to define geographic features, e.g. points, lines, or areas. There is considerable redundancy with this data model since the boundary segment between two polygons can be stored twice, once for each feature. The CAD structure emerged from the development of computer graphics systems without specific considerations of processing geographic features. Accordingly, since features, e.g. polygons, are self-contained and independent, questions about the adjacency of features can be difficult to answer. The CAD vector model lacks the definition of

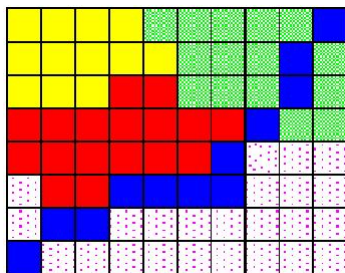
spatial relationships between features that is defined by the topologic data model.

- **Vector Advantages**

- Accurate positional information that is best for storing discrete thematic features (e.g., roads, shorelines, sea-bed features).
- Compact data storage requirements
- Can associate unlimited numbers of attributes with specific features

2. Raster data formats

Raster data models incorporate the use of a grid-cell data structure where the geographic area is divided into cells identified by row and column. This data structure is commonly called raster. While the term raster implies a regularly spaced grid other tessellated data structures do exist in grid based GIS systems. In particular, the quadtree data structure has found some acceptance as an alternative raster data model. The size of cells in a tessellated data structure is selected on the basis of the data accuracy and the resolution needed by the user. There is no explicit coding of geographic coordinates required since that is implicit in the layout of the cells. A raster data structure is in fact a matrix where any coordinate can be quickly calculated if the origin point is known, and the size of the grid cells is known. Since grid-cells can be handled as two-dimensional arrays in computer encoding many analytical operations are easy to program. This makes tessellated data structures a popular choice for many GIS software. Topology is not a relevant concept with tessellated structures since adjacency and connectivity are implicit in the location of a particular cell in the data matrix.



Several tessellated data structures exist, however only two are commonly used in GIS's. The most popular cell structure is the regularly spaced ma-

trix or raster structure. This data structure involves a division of spatial data into regularly spaced cells. Each cell is of the same shape and size. Squares are most commonly utilized.

Since geographic data is rarely distinguished by regularly spaced shapes, cells must be classified as to the most common attribute for the cell. The problem of determining the proper resolution for a particular data layer can be a concern. If one selects too coarse a cell size then data may be overly generalized. If one selects too fine a cell size then too many cells may be created resulting in a large data volume, slower processing times, and a more cumbersome data set. As well, one can imply accuracy greater than that of the original data capture process and this may result in some erroneous results during analysis.

As well, since most data is captured in a vector format, e.g. digitizing, data must be converted to the raster data structure. This is called vector-raster conversion. Most GIS software allows the user to define the raster grid (cell) size for vector-raster conversion. It is imperative that the original scale, e.g. accuracy, of the data be known prior to conversion. The accuracy of the data, often referred to as the resolution, should determine the cell size of the output raster map during conversion.

Most raster based GIS software requires that the raster cell contain only a single discrete value. Accordingly, a data layer, e.g. forest inventory stands, may be broken down into a series of raster maps, each representing an attribute type, e.g. a species map, a height map, a density map, etc. These are often referred to as one attribute maps. This is in contrast to most conventional vector data models that maintain data as multiple attribute maps, e.g. forest inventory polygons linked to a database table containing all attributes as columns. This basic distinction of raster data storage provides the foundation for quantitative analysis techniques.

This is often referred to as raster or map algebra. The use of raster data structures allow for sophisticated mathematical modelling processes while vector based systems are often constrained by the capabilities and language of a relational DBMS.

This difference is the major distinguishing factor between vector and raster based GIS software. It is also important to understand that the selection of

a particular data structure can provide advantages during the analysis stage. For example, the vector data model does not handle continuous data, e.g. elevation, very well while the raster data model is more ideally suited for this type of analysis. Accordingly, the raster structure does not handle linear data analysis, e.g. shortest path, very well while vector systems do. It is important for the user to understand that there are certain advantages and disadvantages to each data model.

- **Raster Advantages**

- It is the most common data format
- It is easy to perform mathematical and overlay operations
- Satellite information is easily incorporated
- Better represents “continuous”- type data

3. Image data

Image data is most often used to represent graphic or pictorial data. The term image inherently reflects a graphic representation, and in the GIS world, differs significantly from raster data. Most often, image data is used to store remotely sensed imagery, e.g. satellite scenes or orthophotos, or ancillary graphics such as photographs, scanned plan documents, etc.

Image data is typically used in GIS systems as background display data (if the image has been rectified and georeferenced); or as a graphic attribute. Remote sensing software makes use of image data for image classification and processing. Typically, this data must be converted into a raster format (and perhaps vector) to be used analytically with the GIS. Image data is typically stored in a variety of de facto industry standard proprietary formats. These often reflect the most popular image processing systems. Other graphic image formats, such as TIFF, GIF, PCX, etc., are used to store ancillary image data. Most GIS software will read such formats and allow you to display this data.



3.4. Attribute data models

A separate data model is used to store and maintain attribute data for GIS software. These data models may exist internally within the GIS software, or may be reflected in external commercial Database Management Software (DBMS). In the raster data model, the cell value (Digital Number) is the attribute. Examples: brightness, land-cover code, SST, etc.

For vector data, attribute records are linked to point, line & polygon features. Can store multiple attributes per feature. Vector features are linked to attributes by a unique feature number. A variety of different data models exist for the storage and management of attribute data. The most common are:

- Tabular
- Hierarchical
- Network
- Relational
- Object Oriented

The tabular model is the manner in which most early GIS software packages stored their attribute data. The next three models are those most commonly implemented in database management systems (DBMS). The object oriented is newer but rapidly gaining in popularity for some applications.

Revision Questions

EXERCISE 5.  Define a map.

Example . Distinguish between vector and raster formats.

Solution: Revise.



EXERCISE 6.  What is a spatial data model?

LESSON 4

Tabular Data Design

Learning outcomes

By the end of this topic you should be able to;

- *Define data modeling.*

4.1. Introduction

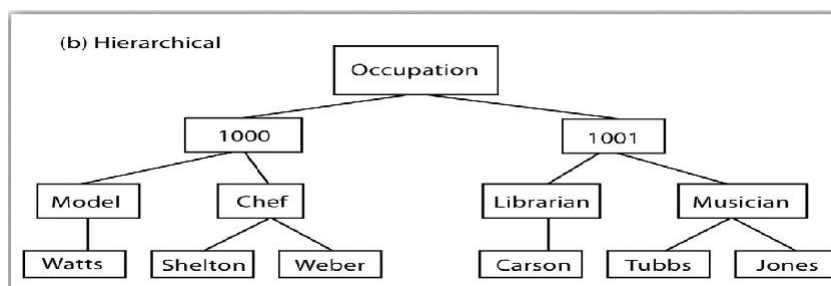
Tables serve as a container for records about entities that can be distinguished by their attributes. Data modeling is the process of defining features to be included in the database, their attributes and relationships, and their internal representation in the database. It involves the development of conceptual, logical and physical models of the census geographic database. The outcomes include a comprehensive data dictionary that defines the content of the databases that are produced by the agency. In some instances, such dictionaries can be adopted or adapted from other agencies in the country, for example, where a national digital topographic database exists. In other cases, this data dictionary has to be developed from scratch. The resources required for this will depend on how comprehensive the database will be. It may also be necessary to integrate existing database models that have been developed to manage tabular census information. This is necessary, for example, if data from previous censuses need to be integrated in the GIS databases. In addition to the data dictionary, a procedural manual defines the steps necessary in the development and processing of digital spatial data. Such manuals are important for ensuring consistency in the products that are generated by different technicians or units that may, perhaps, be scattered across the country. They also define recurring analytical work such as the methods used to reconcile past census data with new boundaries after the administrative units have been changed. Accuracy standards should also be defined as part of the overall database design process. While accuracy is often not a critical issue in census mapping – in fact, many countries rely on hand-drawn sketch maps for this purpose – it becomes an issue when the resulting census maps are used in combination with other, higher-accuracy data.

Several types of database models exist, such as the flat, hierarchical, network, and relational models. A flat database is essentially a spreadsheet whereby all data are stored in a single, large table. A hierarchical database is also a fairly simple model that organizes data into a “one-to-many” association across levels. Common examples of this model include phylogenetic trees for classification of plants and animals and familial genealogical trees showing parent-child relationships. Network databases are similar to hierarchical databases, however, because they also support “many-to-many” relationships. This expanded capability allows greater search flexibility within the dataset and reduces potential redundancy of information. Alternatively, both the hierarchical and network models can become incredibly complex

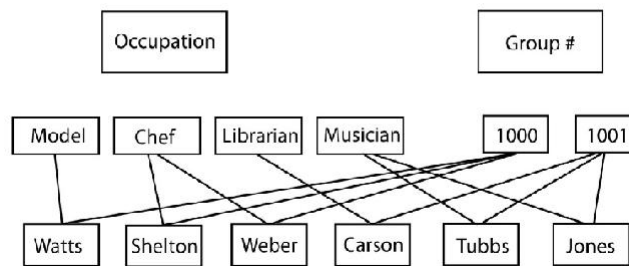
depending on the size of the databases and the number of interactions between the data points. Modern geographic information system (GIS) software typically employs a fourth model referred to as a relational database.

(a) Flat File

Name	Group #	Occupation
Watts	1000	Model
Shelton	1000	Chef
Weber	1000	Chef
Tubbs	1001	Musician
Jones	1001	Musician
Carson	1001	Librarian



(c) Network



4.2. Querying GIS data

1. Attribute query

- Select features using attribute data (e.g. using SQL)
- Results can be mapped or presented in conventional database form
- Can be used to produce maps of subsets of the data or choropleth maps

2. Spatial query is used by clicking on features on the map to find out their attribute values. The combination of these is a powerful way of exploring spatial patterns in your data.

Revision Questions

EXERCISE 7.  What is data modeling?

Example . Discuss the two types of queries.

Solution: Revise.



EXERCISE 8.  Distinguish the different types of database files.

LESSON 5

Data Sources & Data Collection

Learning outcomes

By the end of this topic you should be able to;

- *Identify sources of geographical data*
- *Know geographical data collectors and providers*

5.1. Introduction

Word processors, spreadsheets and graphics packages are programs that let you create and edit digital data. Each type of application saves its data into a particular file format. For example, a graphics program will let you save your drawing as a '.jpg' JPEG image, word processors let you save your document as an '.odt' OpenDocument or '.doc' Word Document, and so on. Just like these other applications, GIS Applications can store their data in files on the computer hard disk. There are a number of different file formats for GIS data, but the most common one is probably the 'shape file'. The name is a little odd in that although we call it a shape file (singular), it actually consists of at least three different files that work together to store your digital vector data. Many GIS Applications are also able to store digital data inside a database. In general, storing GIS data in a database is a good solution because the database can store large amounts of data efficiently and can provide data to the GIS Application quickly. Using a database also allows many people to work with the same vector data layers at the same time. Setting up a database to store GIS data is more complicated than using shapefiles.

Table 1: The basic files that together make up a 'shapefile'.

Extension	Description
.shp	The geometry of vector features are stored in this file.
.dbf	The attributes of vector features are stored in this file.
.shx	This file is an index that helps the GIS Application to find features more quickly.

5.2. Sources of Data

Creating a GIS database is a daunting task. It involves data capture, verification, and structuring processes. Raw geographical data is available in many different analogue and digital forms such as maps, aerial photographs, satellite images, or tables. Creation of a spatial database may involve the following:

- Acquire digital database from a supplier
- Digitize existing analogue data

- Carry out one's survey of geographical entities
- Interpolate from point observations to continuous surfaces

All data must be placed in a common coordinate system and well coded to be stored in the internal database of the GIS, resulting in a current, complete database for subsequent spatial analysis and modelling. As previously identified, two types of data are input into a GIS, spatial and attribute. The data input process is the operation of encoding both types of data into the GIS database formats. The creation of a clean digital database is the most important and time consuming task upon which the usefulness of the GIS depends. The establishment and maintenance of a robust spatial database is the cornerstone of a successful GIS implementation.

As well, the digital data is the most expensive part of the GIS. Yet often, not enough attention is given to the quality of the data or the processes by which they are prepared for automation. The general consensus among the GIS community is that 60 to 80 % of the cost incurred during implementation of GIS technology lies in data acquisition, data compilation and database development.

A wide variety of data sources exist for both spatial and attribute data. The most common general sources for spatial data are:

1. hard copy maps;
2. aerial photographs;
3. remotely-sensed imagery: derived from optical and digital remote sensing systems mounted on aircraft and satellite platforms provide much spatial information over many levels of spatial and temporal resolution
 - Stereo aerial photographs are overlapping analogue images that have many applications.
 - They are a major source of data for human interpretation and mapping of geology, soil, vegetation, or land cover.
 - They are also valuable background documents for placing other spatial data in a geographical context
 - Digital photogrammetry and orthophoto mapping provide data on terrain elevation and land cover directly in digital form.

- Wide range of scanners on satellites provide digital data directly:

Passive systems: sense reflected and emitted radiation

Active systems: side-band radar scanners, laser altimeter scanners, sonar scanners point data samples from surveys;

4. existing digital data files.
5. Traditional survey methods of ground or field surveys aimed at recording sample values at known locations using instruments ranging from questionnaires, soil augers, to automatic chemical probes.

Existing hard copy maps, e.g. sometimes referred to as analogue maps, provide the most popular source for any GIS project.

Potential users should be aware that while there are many private sector firms specializing in providing digital data, federal, provincial and state government agencies are an excellent source of data. Because of the large costs associated with data capture and input, government departments are often the only agencies with financial resources and manpower funding to invest in data compilation. An inherent advantage of digital data from government agencies is its cost. It is typically inexpensive. However, this is often offset by the data's accuracy and quality. Attribute data has an even wider variety of data sources. Any textual or tabular data that can be referenced to a geographic feature, e.g. a point, line, or area, can be input into a GIS. Attribute data is usually input by manual keying or via a bulk loading utility of the DBMS software. ASCII format is a de facto standard for the transfer and conversion of attribute information.

It is important that all spatial data in a GIS is located with respect to a common frame. Placing these in a common system in GIS is attained through georeferencing. This frame is provided in one of a number of geodetic coordinate systems:

- Plane Orthogonal Cartesian Coordinates oriented conventionally north-south and east-west and derived from geographical coordinates
- Geographical (spherical) coordinates on ellipsoids for mapping the true curved surface of the earth

5.3. Geographic data collectors and providers

5.3.1. Systematic data collectors

- Government agencies: national and regional mapping agencies
 - Have the mandate to define and maintain the map spheroids, projections, grid origins, orientation and spacing, reference sea levels, data measurements and collection standards
 - Carry out cadastral surveys and registration of utilities in a systematic manner
 - Systematic collection of remote sensing data is carried out by government and private remote sensing corporations in some countries such as Canada, USA, Russia, India, France and Japan

5.3.2. Ad hoc Data Collectors

- Data collected by private surveyors, commercial organizations, market researchers, political organizations, or academic institutions
- Data are for specific purposes
- Access to these data is often restricted because of commercial interests

5.3.3. Data providers

- Offer and sell geographical data in a variety of detail, format, scales and structures.
- Sell data that they have processed but may not have acquired e.g. Regional Centre for Services in Surveying and Mapping in Kenya.
- Some data is provided free e.g. elevations, roads, towns, rivers.

5.4. Acquiring digital datasets

A user should consider the following characteristics of data to ensure they are compatible with the application:

- The currency of the data,
- The length of the record,

- Scale of the data,
- Georeferencing system used
- Data collection technique and sampling strategy
- The quality of data collected
- Data classification and interpolation methods used
- Size and shape of the individual mapping units

After compatibility, the next issue is transfer of the data from the source to the GIS

- Encoding from source format to target format may be necessary
- Standards of interoperability have been developed , systems reading other systems' format
- Remote sensing data may need pre-processing before it can be input to a GIS i.e. Resolution adjustment, cartographic projection, classification, raster-vector conversion etc.

Data may also be gotten online with the Internet or Intranet

5.5. Creating digital datasets by manual input

Manual input of data into a GIS involves four (4) main stages:

1. **Entering the spatial data:** Geographical data in entity model are in terms of points, lines, or areas/pixels defined by a series of coordinates. Its simplest way is to type the coordinates into a file or input program of a GIS. This has been reduced by use of digitizers, scanners, or stereoplotters to encode the X and Y coordinates of desired points
2. **Entering the attribute data:** Attribute data are non-spatial characteristics of the spatial entity that need to be handled by GIS. Few attributes may be added at input of the spatial entity. Others come from different sources such as paper records, existing databases, spread sheets. They may be input manually, or imported into the GIS using a standard transfer, if in a relational database, an identifier is included in the attribute record to link the spatial and attribute data

3. **Spatial data and attribute verification and editing:** It is important to check for errors, inaccuracies, omissions, etc. prior to linking the spatial and attribute data. The best way to check for errors is to produce a computer plot or print preferably on a translucent or thin paper and visually compare with the original print. Certain GIS show topological or identifier errors directly by colour coding the screen image. Attributes may be visually checked by printing out files or by eye; computer programs can be used to scan for gross errors. Errors that may arise during capture of spatial and attribute data may be grouped as follows:

- Spatial data are incomplete or double
- Spatial data in the wrong place
- Spatial data defined using too many coordinates
- Spatial data at the wrong scale: occurs during georeferencing
- Spatial data are distorted: base maps not in correct scale

Typical errors arising in digitizing data are

- Dangling chain: (undershoot or overshoot)
 - Unlabelled polygon: due to duplicate digitizing of a single chain
 - Conflicting labels for a polygon: in case of multiple seed points for a given polygon, e.g. a group of islands belonging to a single polygon
 - Line segments with the same left and right labels; due to mislabelling of adjacent polygons
 - * The errors may be addressed through the editing and updating functions supported directly by most GIS.
 - * Data editing is usually undertaken by viewing the portion of the map containing errors on the screen and correcting them through the software using the keyboard, screen cursor controlled by mouse, or a small digitizer tablet
 - * Minor location errors can be corrected by moving the spatial entity through the screen cursor or by entering a new location on the digitizing tablet

- * Data scaling problems can be overcome by applying a simple numerical functions, e.g. through georeferencing
- * Excess coordinates can be removed through “weeding algorithms, with visual appearance being improved using splines.
- * Attribute values and spatial errors in raster are edited by changing values of the faulty cell followed by inputting the correct attribute value.

4. **Linking the spatial to the attribute data:** It is available in GIS with relational database structures and attained by use of identifiers which are common both in the spatial record and the attribute record. Identifiers are either automatically generated or added manually during digitizing or scanning/ vectorization process. Polygon identifiers are usually added after generation of topology. After the data capture process, we have geometrically correct raster data, or topologically and geometrically correct vector data. Most GIS accept both kinds of data. Structuring of the database as a whole is essential to optimize on storage requirement and performance of the GIS through sensible ordering and data compression techniques.

Revision Questions

EXERCISE 9.  Differentiate between passive and active systems.

Example . Identify errors in data files.

Solution: Revise.



EXERCISE 10.  Describe the stages of creating digital datasets.

LESSON 6

Spatial Data Files

Learning outcomes

By the end of this topic you should be able to;

- *Explain spatial data files*
- *Describe the presentation of the data files*

6.1. Introduction

Spatial data files are like other files you work with on a computer.

6.2. Characteristics of spatial data files

1. Spatial data files can be:
 - stored on a hard drive, memory stick, CD, DVD
 - assigned either a user-defined file name, or are given default file name by a software application
 - organized into folders
 - have the ability to be opened, viewed and edited by one or more GIS software applications that understand their format.
2. However, that is where the similarities end. Spatial data files are unique in that they store “georeferenced” information – information that defines location or place. In addition, descriptive information about the georeferenced information is stored in each spatial data file.
3. Thus rather than just text or numbers, an individual spatial data file is a digital representation of a similar group of geographic features on the surface of the earth.
4. The geographic features can be actual physical entities or events, or they can represent conceptual features.
5. Examples of individual spatial data files representing real geographic features or events are lakes, rivers, wetlands, elevation contours, roads, forested areas, rare species habitats, soils, earthquakes, vehicle thefts, electricity distribution lines, and groundwater reservoirs.
6. Examples of individual spatial data files representing conceptual geographic features are census tract boundaries, zoning boundaries, or parcel boundaries i.e. conceptual features do not physically exist on the landscape, but are imposed by us for various reasons and can be represented in a geographic context.

7. Each spatial data file is uniquely constructed to work within GIS software applications and consists of unique characteristics:

- “shapes” that attempt to reflect / convey the appearance and position of individual geographic features as accurately as possible
- records within a related tabular database that contain numeric and/or textual descriptions of each feature
- a coordinate system that defines the true location of all the features on the earth’s surface (i.e. the latitude/longitude)

6.3. Formats of Spatial Data Files

Spatial data files come in several different formats. You may need to use only one, or you may need to use a combination of them, depending on your particular application and/or type of analysis. Each format falls under one of two different categories: vector or raster.

- Vector spatial data files are ones in which the geographic features being represented are built by a collection of vertices and lines.
- Raster spatial data files are ones in which the geographic features across an entire area are represented by a continuous set of “pixels” or “cells”.

6.3.1. The Shapefile spatial data file format

This is a very common format for spatial data files in the vector category. In this format, geographic features can be represented in one of three ways:

- points
- lines
- polygons

If you utilize an existing shapefile, the point, line or polygon representation is chosen by the organization responsible for its development. If you create your own shapefile, you will have to determine which representation is best for your work or application. The determination is made based on several factors, including but not limited to:

- the need to depict features at a specific scale
- the need to depict features as realistically as possible
- the need to quantify some aspect of geographic features.

6.3.2. The Coverage spatial data file format

This was the original spatial data file format used in GIS software. While this format has taken a “backseat” to the shapefile format due to the simplicity of shapefiles, coverages are still very viable and have many advantages. Along with shapefiles, the coverage format is being superseded by the new geodatabase format. However, many web sites still offer spatial data for download in the coverage format. Just like shapefiles, geographic features are represented as points, lines, or polygons and many factors come into play when deciding which representation is best. Coverages also fall within the vector category. Unlike shapefiles, a single coverage is actually comprised of two folders. Each folder contains a multitude of other files that the GIS software “puts together” in order to represent geographic features and associated tabular information when it is opened. If either folder is missing, or if files from within either folder are missing, the coverage will be “corrupt” and not useable.

6.3.3. The Grid spatial data file format

In most respects, grids are very different from either shapefiles or coverages. Grids fall into the raster category; they are constructed of rows and columns of pixels instead of vertices and arcs. Like coverages however, grids are comprised of two folders, each containing files that the software “puts together” for display and manipulation. Grids can be either:

- Integer Grids: The pixel values are integers and each integer may also be associated with one or more textual descriptions.
- Floating Point Grids: The pixel values will be expressed as decimals. Floating point grids can not have textual descriptions.

6.3.4. Images as Spatial Data Files

Many different image formats can be used in GIS. Some of these may be more familiar than others: .jpg, .tif, .bil, .png, .img, .sid. All image formats fall within the raster category of spatial data. In some cases, images are not used specifically as “spatial data”, but are used to enhance spatial data by providing a digital photograph of a place or object.

In other cases, the images themselves are spatial data. Data provided from the Landsat satellite is an example of imagery that is spatial. If you have ever used Google Earth, the images that appear when you zoom in are spatial data. When an image is “georeferenced” - meaning that information is embedded within the image that describes its position on the surface of the earth in real world coordinates (latitude/longitude), it becomes spatial data. In addition to being “georeferenced”, many images may also be “orthorectified”. This term refers to a complex process wherein distortions caused by differences in terrain elevation, camera tilt, and edge effects are removed from the image. Images that are both georeferenced and orthorectified are frequently called “orthophotographs” or just “orthos” in short.

6.3.5. Computer-Aided Drafting (CAD) files

CAD software applications such as AutoCAD and MicroStation produce vector data in .dwg or .dxf format. ArcGIS software can read these files directly, they do not need to be imported or changed in any way. In most instances, but not all, these .dwg or .dxf files will include georeferencing information and thus, the data are spatial. For those files that are not georeferenced, there are methods and / or separate software programs that allow them to become georeferenced. CAD files often provide very detailed information such as the floor plan of a building which can be useful for certain GIS applications.

6.3.6. The GeoDataBase (GDB)

The GDB is the newest spatial data file format developed by ESRI. It is a replacement file format for all of the above. GDBs have very specific advantages over the other file types:

1. All spatial data resides within a single file. This eliminates having several different file formats residing in several different folders across a disk. The

file has an .mdb extension i.e. a GDB containing all spatial data for Nairobi might be named nairobi.mdb

2. GDBs have greater “intelligence”
3. Industry specific data models can be applied to a GDB

6.4. Naming conventions for Spatial Data Files & Folders containing Spatial Data Files

Since spatial data files have a complex structure, it is extremely important that you follow some basic rules for assigning names to these files as well as to the folders that contain these files. Thus, to avoid these problems from ever happening to you, follow these simple rules:

- Keep the names short
 - Between 8 and 15 characters
 - If you work with spatial data that are in the “coverage” vector format, or the “grid” raster format, only 8 characters are allowed.
- Do Not Use Spaces
 - town boundaries is a bad name for a file; use town_boundaries instead
 - my gis data is a bad name for a folder that will contain spatial data files
- Do Not Use Special Characters (the only exception is the underscore)
 - e.g. town#boundaries is a bad name for a file
 - e.g. my!gis!data is a bad name for a folder
- Never Put A Number As The First Character
 - e.g. 2000population is a bad name for a file
 - a good alternative would be y2000pop
- Keep The Names Simple

- use abbreviations where possible
 - use underscores where possible as a replacement for spaces
 - use upper case where it makes sense
 - e.g. twn_bndy or TwnBndy
- Implement A Tracking Method
 - When you begin to perform analysis, this rule will make more sense than it probably does now but it is important to keep in mind.
 - When you manipulate vector spatial data files, you will frequently be generating a copies of the data that have “added value”.
 - When you manipulate raster data files, you will frequently be performing a process multiple times before you get the outcome you wish.
 - Therefore, it is important that you have a naming convention that will allow you to keep track of these sequential files. Example, you may start with a shapefile of town_boundaries which only has one attribute (e.g. the name of the town). Performing a process called a Join, results in new attributes (population of each town, zip code, the number of grocery stores in each town, etc) being appended to the table. Once the join is completed, you would export the entire file to a new file in order to permanently retain the new attributes. Because you have implemented a tracking method it will be easy to distinguish the latest file you are working with.

Revision Questions

EXERCISE 11.  Define georeferencing.

Example  Distinguish between vector and raster.

Solution: Revise.



EXERCISE 12.  List three rules to follow in creating data files.

LESSON 7

Cartographic Design

Learning outcomes

By the end of this topic you should be able to;

- *Identify security threats to databases*
- *Discuss multi-level database security*

7.1. Introduction

7.2. Principles of cartographic design

1. Use of color: The primary function of color is to make information on a map visually distinguishable

Dimensions of color

- Hue: dominant wavelength (i.e., color) - Hue is generally used to indicate qualitative (nominal) differences across the map area
- Value: how light or dark a given hue is
- Saturation: purity of hue (range of reflected wavelengths - Value and saturation are typically used to represent quantitative (ordinal, interval, or ratio) differences across the map area

2. Use of text: Descriptive text is used to give a map its title, to explain the legends and label features. Text is stored as either symbols of a coverage or annotation coverage

Text characteristics:

- Family Arial vs Helvetica
- Style (face) bold vs italic
- Font 32 point vs 48 point
- Color black vs blue

3. Symbols and symbol sets: A symbol is a graphic pattern that is used to represent a feature on a map.

- Marker symbol: points and nodes (.mrk)
- Line symbols: arcs (.lin)
- Shade symbols: polygons (.shd)
- Text: annotation (.txt)

In symbol sets, each symbol must be programmed individually, so GIS software packages usually supply the user with predefined symbol sets that can be edited and customized.

4. Map-to-page transformation

- Physical page: actual surface of display medium
- Graphics page: portion of physical page where map is drawn
- Map limits: portion of graphics page where coverage features are drawn
- Map extent: area of the earth's surface to be displayed (in actual ground units)

Map composition

- Map layout design: cartography is both a science and an art, therefore subjectivity and creativity play an important role in the aesthetic quality of a map
- Geographical contents: elements to be included are governed by:
 - Map theme i.e., land use
 - Map coverage i.e. Illinois vs. U.S.
 - Map scale i.e. level of generalization
- Label placement: Good label placement ensures readability and enables the reader to associate labels with the map elements that they describe.

7.2.1. Geographic contents to be considered:

- Visual balance: location/proportionality
- Visual clarity: generalization
- Visual hierarchy: font & symbol size
- Visual contrast: use of color and shading
- Context: selection of data layers
- Text: annotation of features

7.2.2. Cartographic generalization

1. Why? When geographic data are gathered at a scale that is larger than the scale at which they are presented, it is necessary to reduce the complexity of the data to make the resulting map more aesthetically pleasing. Generalization ensures that geographic data are presented at a scale appropriate to the purpose of the map and the application requirements of the user.

2. When?

- Simplification
- Smoothing
- Aggregation
- Amalgamation
- Merging
- Collapse
- Refinement
- Typification
- Exaggeration
- Displacement
- Classification

3. How?

- **Congestion:** too many features in too little space
- **Coalescence:** features touch due to inadequate symbolization
- **Conflict:** feature symbol incompatible with background
- **Complication:** data from different sources or at different scales or levels of tolerance
- **Inconsistency:** Generalization applied in a nonuniform fashion across map
- **Imperceptibility:** Feature falls below minimum resolution of map

EXERCISE 13.  Explain the advantages of role base access control in multi level databases.

Example . Giving suitable examples, differentiate between credentials and population as used in multilevel databases.

Solution: Apulation or the average income in the population. □

EXERCISE 14.  Explain covert channel as used in multilevel data bases.

LESSON 8

Basics of Remote Sensing

Learning outcomes

By the end of this topic you should be able to;

- *Explain the functions of a firewall*
- *Explain the limitations of firewalls*
- *Describe various types of firewalls*

8.1. Introduction

No article on GIS would be complete without mentioning some of the applications of spatial data. One important application is wildlife habitat assessment. Spatial data can be used to calculate the home range or territory of a particular species. They can also be used to eliminate areas where a certain animal would rarely or never be found (e.g., slope data could be used to eliminate extremely steep slopes, or elevation data could be used to eliminate elevations above a certain level). In addition, by simply moving a 3 x 3 pixel window over the vegetation map, calculations

of juxtaposition and interspersions can be made. Juxtaposition can measure the number of different vegetation types in a certain area, while interspersions measures the value of the edges. Both are very important to wildlife, since food and cover must be in close proximity in order for the habitat to be useful. Obviously, other proximity analyses are possible, such as distances to water and roads. A wildlife habitat assessment model, typically based on knowledge compiled by one or more wildlife specialists, combines the spatial data, its derivatives, and other information necessary to predict habitat quality. The more important variables are frequently given higher weights. This analysis can determine areas of good, marginal, and poor habitat, providing the first step toward intelligently improving the area.

Another application, predicting snowmelt runoff from a watershed, involves a statistical model using regression analysis. As in the wildlife example, the same spatial data are useful. However, in this case the regression analysis chooses the important variables instead of the wildlife biologist. In predicting snowmelt runoff the important parameters are the snow-covered area and the snow-water equivalent. Other spatial data could include elevation, slope, aspect, vegetation, thermal emittance, near-infrared reflectance, and soils. In addition, data must be collected at specific sample locations on the ground. All these data are tested by the regression analysis to see which are significant in predicting runoff; the best model has the highest correlation coefficient.

This model is then used to predict snowmelt runoff over the entire watershed. These two examples demonstrated different modeling approaches using spatial data. It should not be inferred from these examples that GIS only applies to natural resources. GIS has many applications, from aiding the census to evaluating polluted owl habitat. Those starting to use spatial data as a tool for decisionmaking should

review the literature in their particular field and learn as much as possible from past successes and failures. Finally, a warning. A GIS is a very powerful decision-making tool. However, it is just a tool. There is a great temptation to lean too heavily on the computer and to stop thinking. There is also a tendency to collect too much spatial data. One more layer of data is not always the answer. Be thoughtful and wary, and GIS will serve you well.

8.2. GIS Applications

- Site selection
 - Helicopter Landing Zones
 - Amphibious Assault (Water Depth)
 - Buffer Zones
 - Flight Planning
 - Battlefield Visualisation
- Cross country movement
 - Route planning
 - Intervisibility study
- Facilities management
- Airfield assessment
- Road network analysis (convoys)
- Propagation coverages
- Observation post siting analysis
- Perspective views

Revision Questions

EXERCISE 15.  Explain security attacks that can be implemented on packet-filtering routers & countermeasures.

Example . List the characteristics of bastion host firewall.

Solution: a) execution file i) have each proxy run as a non-privileged user in a private and secured directory. □

EXERCISE 16.  Explain the feature of a distributed firewall that distinguishes it from other types of firewalls.

LESSON 9

Implementation Issues & Future of GIS

Learning outcomes

By the end of this topic you should be able to;

- *Define multilevel security*
- *Explain trusted systems*

9.1. Current options

Perhaps the first question asked by anyone when discovering GIS is what are the current options available ?. This question is often asked as directly as what is the best GIS ?. Quite simply, there is no best GIS. A wide variety of GIS software offerings exist in the commercial market place. Commercial surveys often are a good starting point in the assessment of GIS software. The number of GIS software offerings is approximately 10 if one eliminates the following:

- the university based research software, which tends to lack full integration and usually has narrow channels of functionality;
- the CAD vendors, who like to use GIS jargon but often cannot provide full featured functionality; and the consulting firms, that will provide or customize selected modules for a GIS but lack a complete product.
- One of the problems in evaluating the functionality of GIS software is the bias one gets from using one system or another. Comparing similar functions between systems is often confusing.
- Like any software, ultimately some do particular tasks better than others, and also some lack functionality compared to others.
- Any GIS should be evaluated strictly in terms of the potential user's needs and requirements in consideration of their work procedures, production requirements, and organizational context !
- The experienced GIS consultant can play a large and valuable role in the assessment process.
- A current accepted approach to selecting the appropriate GIS involves establishing a
- benchmark utilizing real data that best represents the normal workflow and processes employed in your organization.
- Development of the benchmark should include a consideration of other roles within your organization that may require integration with the GIS technology. A logical and systematic approach as such is consistent with existing

information systems (IS) planning methodologies and will ultimately provide a mechanism for a successful evaluation process.

9.2. Justification & expectations

GIS is a long term investment that matures over time. The turnaround for results may be longer term than initially expected. Quite simply, GIS has a steep learning curve. The realization of positive results and benefits will be not achieved overnight. Even though the proper assessment of an appropriate GIS product requires a good understanding of user's needs, most often systems are acquired based on less than complete

and biased evaluations. Nonetheless, even with the GIS in hand a properly structured and systematic implementation plan is required for a successful operation. Generally, a GIS implementation plan must address the following technical, financial, and institutional considerations:

- system acquisition tactics and costs;
- data requirements and costs;
- database design;
- initial data loading requirements and costs;
- system installation tactics, timetable, and costs;
- system life cycle and replacement costs;
- day-to-day operating procedures and costs;
- staffing requirements and costs;
- user training and costs; and
- application development and costs.

Potential GIS buyers should be aware of the necessary investment required in hardware, software, training, supplies, and staffing. The cost of establishing a successful GIS operation is substantial. However, with realistic expectations and support the

development of GIS within an organization that manipulates geographic data will almost certainly prove beneficial.

Certain considerations of data longevity, data capture, personnel hiring, etc. are the practical concerns of GIS implementation. The longer term implications, such as hardware/software maintenance and replacement, should also be considered. The acquisition of GIS technology should not be done without seriously considering the way in which GIS will interact with the rest of the organization.

9.3. Implementation Issues

The mere presence of an implementation plan does not guarantee success. Most organizations do not have sufficient staff to cope with the commitment and extra work required when introducing a GIS to existing operations. GIS implementation must also consider all technology transfer processes.

9.3.1. Common Pitfalls

Several pitfalls exist that most often contribute to the failure of a GIS implementation strategy. These are identified below:

- Failure to identify and involve all users Users in an operational GIS environment consist of operations, management, and policy levels of the organization. All three levels should be considered when identifying the needs of your users.
- Failure to match GIS capability and needs. A wide spectrum of GIS hardware and software choices currently exist. The buyer is presented with a significant challenge making the right choice. Remember, the right choice will be the GIS that provides the needed performance no more, no less for the minimum investment. The success of a GIS implementation is particularly sensitive to the right hardware and software choices
- Failure to identify total costs. The GIS acquisition cost is relatively easy to identify. However, it will represent a very small fraction of the total cost of implementing a GIS. Ongoing costs are substantial and include hardware and software maintenance, staffing, system administration, initial data loading, data updating, custom programming, and consulting fees.

- Failure to conduct a pilot study The GIS implementation plan concerns itself with the many technical and administrative issues and their related cost impacts. Three of the most crucial issues, are database design, data loading and maintenance, and day-to-day operations. The pilot study will allow you to gather detailed observations, provided it is properly designed, to allow you to effectively estimate the operational requirements.
- Giving the GIS implementation responsibility to the EDP Department. Because of the distinct differences of the GIS from conventional EDP systems, the GIS implementation team is best staffed by non-data processing types. The specialized skills of the 'GIS analyst' are required at this stage. Reliance on conventional EDP personnel who lack these skills will ensure failure.
- Failure to consider technology transfer. Training and support for on-going learning, for in-house staff as well as new personnel, is essential for a successful implementation. Staff at the three levels should be educated with respect to the role of the GIS in the organization. Education and knowledge of the GIS can only be obtained through on-going learning exercises. Nothing can replace the investment of hands on time with a GIS

A critical requirement for all GIS implementations is that adequate education and training is provided for operational staff, as well as realistic priorities are defined with which to learn and apply the technology. This is where a formal training curriculum is required to ensure that time is dedicated to learning the technology properly. Adding GIS activities to a staff member's responsibilities without establishing well defined milestones and providing adequate time and training mechanisms is prone to failure. A focused and properly trained operations staff that has consistent training will result in greatly reduced turnaround times for operations, and ensure consistency in quality of product. The threshold point of the learning curve is typically around the two year time frame.

However, this is dependent on the ability of the organization to establish a well defined and a GIS acquisition based on well defined user needs and priorities is more likely to succeed than without. A major pitfall of most installations with GIS technology, e.g. particularly forestry companies and government agencies, is the lack of well defined user needs on which to base the GIS acquisition and implementation.

9.4. The Implementation Plan

GIS implementation requires a plan, and in each plan the following issue must be dealt with.

1. The implementation planning process: - organizing the tasks associated with implementation and smoothly making the transition from strategic planning to GIS implementation;
2. Creating an implementation management framework – Assigning roles and responsibilities to project participants.
3. Developing a conceptual overview: - Articulating the concepts envisioned at the awareness stage, a set of discrete parts that can be analysed, specified and implemented;
4. Managing expectations, establishing procedures and establishing an appropriate sequence of events; communicating the requirements for implementation to project participants in enough detail for all to understand what is to be done, how long it will take, what it will cost, and who will need to be involved and for how long.
5. Preparing the implementation plan: documenting what has been determined and agreed upon.
6. Developing “error free” specifications structuring the planning and design process to produce the fewest number of analysis and design errors.
7. Implementing the plan: successfully starting up and carrying through to completion.

9.5. Implementation Process

Implementation can be seen as a six phase process. They are:

1. Creating an awareness

GIS needs to be sold within an organization. The education of staff is very important. Depending on the way in which GIS technology is being introduced to the organization the process for creating an awareness may differ.

Technical workshops are often appropriate when a top-down approach exists, while management workshops are often more relevant when a bottoms-up approach exists. Education of the new technology should focus on identifying existing problems within an organization. These often help justify a GIS acquisition. They include :

- spatial information is poorly maintained or out of date;
 - spatial data is not recorded or stored in a standard way;
 - spatial data may not be defined in a consistent manner, e.g. different classifications for timber information;
 - data is not shared between departments within an organization;
 - data retrieval and manipulation capabilities are inadequate to meet existing needs; and
 - new demands are made on the organization that cannot be met with existing information systems.

2. Identifying System Requirements: The definition of system requirements is usually done in a user needs analysis. A user needs analysis identifies users of a system and all information products required by those users. Often a prioritization of the information products and the data requirements of those products is also undertaken. A proper user needs analysis is crucial to the successful evaluation of GIS software alternatives.

After user needs have been identified and prioritized they must be translated into functional requirements. Ideally, the functional requirements definition will result in a set of processing functions, system capabilities, and hardware requirements, e.g. data storage, performance. Experienced GIS consultants often play a major role in this phase.

3. System Evaluations: Evaluating alternative hardware and software solutions is normally conducted in several stages. Initially a number of candidate systems are identified. Information to support this process is acquired through demonstrations, vendor literature, etc. A short listing of candidates normally occurs based on a low level assessment. This followed by a high level assessment based on the functional requirements identified in the previous phase. This often results in a rating matrix or template. The assessment should take

into account production priorities and their appropriate functional translation. After systems have been evaluated based on functional requirements a short list is prepared for those vendors deemed suitable.

A user-friendly system would include the following:

- System commands easily understood by the non-computer specialist;
- Efficient interaction for expert user. (command lines for faster interaction)
- Error messages should state clearly the nature of the error and probable its cause and direction to rectify the problem.
- A context sensitive 'help' facility at this stage

The second phase; standard benchmark, is then used to determine the system of choice. This is the administration of a standardized test procedure which provides a systematic means of comparing the performance level of competing systems. The buyer assesses how a system will perform under expected operating conditions. This operation starts with data that the vendor has not used and will actually be used in the GIS. A good system should perform those tasks specified as indicated without a specialist to change it along. A pilot study with an organizations data should be carried out to test the robustness of a system. This pilot study provides the necessary training for the staff and experience on the challenge of implementing a system. Potential benefits may also be realized at this stage. The pilot strictly should be carried out by the staff of the organization together with the consultations.

4. **Justifying the System Acquisition:** The proper justification of the chosen system requires consideration of several factors. Typically a cost-benefit analysis is undertaken to analyze the expected costs and benefits of acquiring a system. To proceed further with acquisition the GIS should provide considerable benefits over expected costs. It is important that the identification of intangible benefits also be considered.

The justification process should also include an evaluation of other requirements. These include data base development requirements, e.g. existing data

versus new data needs and associated costs; technological needs, e.g. maintenance, training, and organizational requirements, e.g. new staff, reclassification of existing job descriptions for those staff who will use the GIS.

Some of the benefits of a GIS are:

- Better storage and updating of data
- More efficient information retrieval
- More efficient production of information products
- Rapid analysis of alternatives
- The value of better decisions

5. **System Acquisition and Start Up:** After the system, e.g. hardware, software, and data, is acquired the start up phase begins. This phase should include pilot projects. Pilot projects are a valuable means of assessing progress and identifying problems early, before significant resources have been wasted. Also, because of the costs associated with implementing a GIS it is often appropriate to generate some results quickly to appease management. First impressions are often long remembered.

6. **Operational Phase:** The operational phase of a GIS implementation involves the on-going maintenance, application, and development of the GIS. The issue of responsibility for the system and liability is critical. It is important that appropriate security and transaction control mechanisms be in place to support the system. A systematic approach to system management, e.g. hardware, software, and data, is essential.

Revision Questions

EXERCISE 17.  List the services of trusted computing.

Example . Explain 5 components of trusted platform module.

Solution: i) Cryptographic keys, and session information. □

EXERCISE 18.  What is a trusted computer system?

Solutions to Exercises

- Exercise 1.** Revise. Exercise 1
- Exercise 2.** *Revise.* Exercise 2
- Exercise 3.** Revise. Exercise 3
- Exercise 4.** Revise. Exercise 4
- Exercise 5.** Revise. Exercise 5
- Exercise 6.** Revise. Exercise 6
- Exercise 7.** Revise. Exercise 7
- Exercise 8.** Revise. Exercise 8
- Exercise 9.** Revise. Exercise 9
- Exercise 10.** Revise. Exercise 10
- Exercise 11.** Revise. Exercise 11
- Exercise 12.** Revise. Exercise 12
- Exercise 13.** RBAC ensures that only authorized users arations. Exercise 13
- Exercise 14.** A covert chents or processes. Exercise 14
- Exercise 15. IP address spoofing:** where intruder transmits packets from the outside with interimum fragment size to include full header. Exercise 15
- Exercise 16.** An important aspect of a distributed firewall configuration is security monitoring. Such monitoring typically includes log aggregation and analysis, firewall statistics, and fine-grained remote monitoring of individual hosts if needed. Exercise 16
- Exercise 17.** i) Authentication ii) Boot certification iii) Encryption Exercise 17
- Exercise 18.** A trusted computer system is a system that employs sufficient hardware and software assurance measures to allow its use for simultaneous processing of a range of sensitive or classified information. Exercise 18